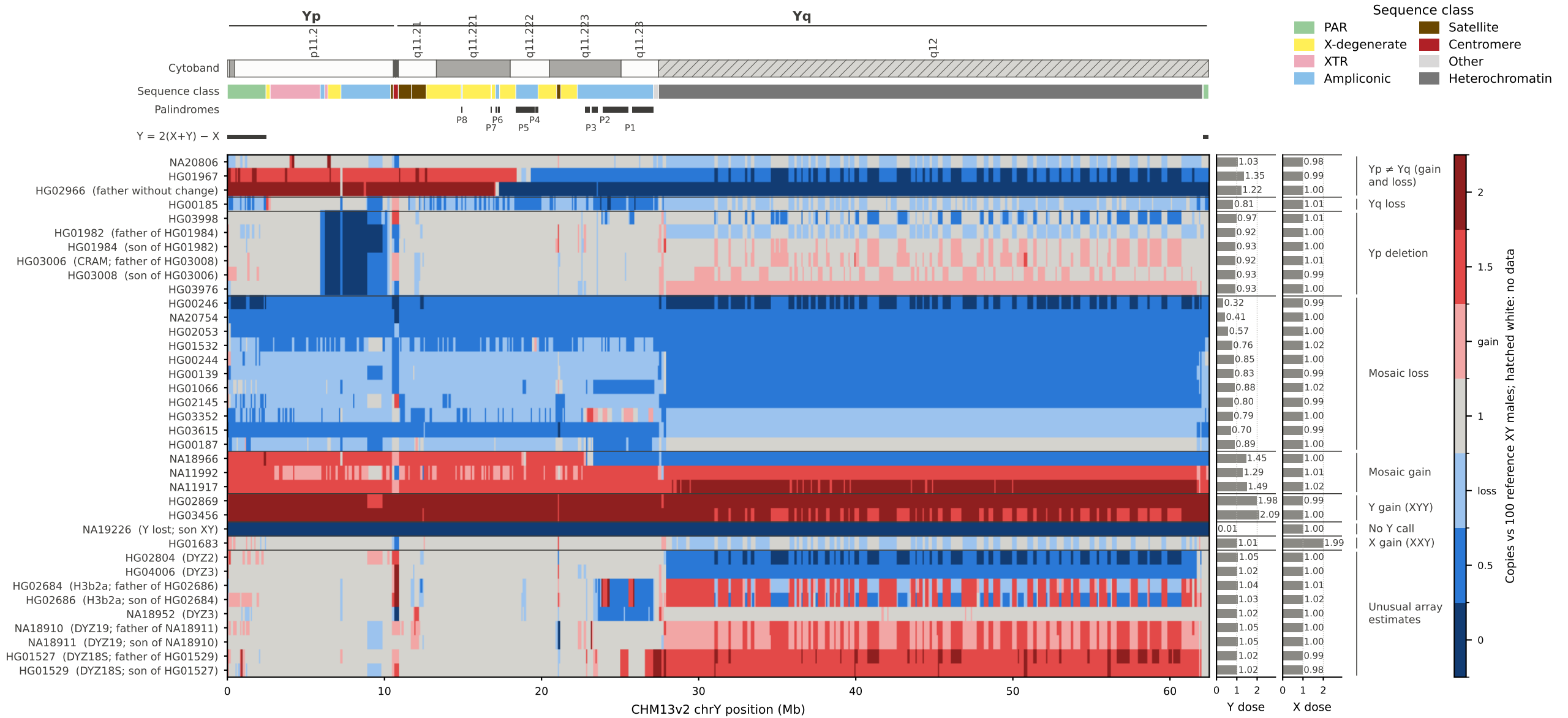

Heatmap: each 100 kb window of CHM13v2 chrY shows the median dose of the CHM13v2 chrY 31-mers absent from every other CHM13 sequence (Y-specific, multi-copy ones included) that lie in the window, counted with yruler in the reads that carry a yruler marker k-mer (picked without alignment), over the median of 100 reference XY males (Y state normal; 1 = as in the reference; 2 = twice). In the centromere (DYZ3), DYZ19 and Yq12 the value is the array amount relative to the median male and depends on the Y lineage (centromere in the 100 reference XY males: 0.156-2.897), not a chromosome copy number. Colour steps: 0 (< 0.25), 0.5 (0.25-0.75), loss (0.75-0.9), 1 (0.9-1.1), gain (1.1-1.25), 1.5 (1.25-1.75), 2 (>= 1.75). Hatched white: fewer than 50 k-mer positions in the window (segmental duplications). HG03006: Y from the 31-mer counts of the chrX, chrY, chromosome 2 and unmapped reads of the CRAM (saved reads written before the whole-Y k-mers were added).
Y dose = 2 x mean count of the Y single-copy ruler k-mers / mean count of the chromosome 2 k-mers (Y against an autosome). X dose = median dose of CHM13 X-specific 31-mers (absent from chrY, autosomes and chrM; 1 in 32 kept), counted with yruler in the chrX, chrY, chromosome 2 and unmapped reads of the CRAM, over the median of 100 reference XY males. PAR1 and PAR2 (black bar under the palindromes): Y dose inferred from CHM13 31-mers present on chrX and chrY and absent from autosomes, Y = 2 x (X + Y dose) - X dose, same reads and reference; a copy change on the X copy of these regions shows as a Y change.
Y state, first rule that applies: no Y call, Y dose <= 0.0222 (highest of 65 heldout females); Yp deletion, Yp dose < 0.5 and Yq dose within 0.1 of 1; Yq loss, Yp dose within 0.1 of 1 and Yq dose more than 0.1 below Yp; Yp≠ Yq (state partial in the tables), Yp and Yq doses differ by more than 0.1, so part of the Y is gained and part lost; Y gain, Y dose >= 1.75; mosaic gain, 1.25 to 1.75; mosaic loss, < 0.9. XYY and XXY need both doses: XYY is Y gain with X dose 1, XXY is X dose 2 with Y dose 1 (k-mer X dose in the 100 reference XY males: 0.981-1.012). Unusual array estimates: males noted in the cohort analysis. Rows are ordered by average-linkage clustering within each group; fathers and sons that are both in the figure are drawn next to each other.